

Solution *Sketches* to Problem Set 2**Problem 1 (Construct Regular Expressions)****10 pts**

Construct regular expressions for the following languages over the alphabet $\Sigma = \{0, 1\}$.

1. $L = \{w : w \text{ contains an odd number of } 1s\}$.

Answer (e.g.): $(0^*10^*1)^*0^*10^*$

2. $L = \{w : w \text{ does not contain the substring } 01\}$.

Answer (e.g.): 1^*0^*

3. $L = \{w : w \text{ has length } \geq 1 \text{ and every even position of } w \text{ contains the symbol } 1\}$. // The first symbol of w is in position 1.

Answer (e.g.): $(\Sigma 1)^*\Sigma \cup (\Sigma 1)^+$

4. $L = \{w : w \text{ the length of } w \text{ is at most } 5\}$.

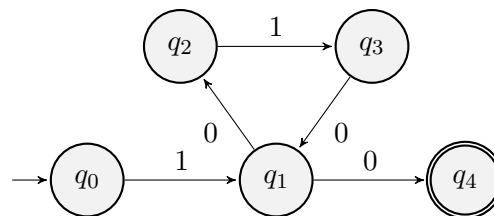
Answer (e.g.): $(\epsilon \cup \Sigma)^5 \text{ or } \epsilon \cup \Sigma \cup \Sigma^2 \cup \Sigma^3 \cup \Sigma^4 \cup \Sigma^5$.

Problem 2 (Regular Expressions to NFA)**10 pts**

For each of the languages given by the regular expressions below, give an example of a string that is in the language, and a string that is not in the language. Then construct an NFA that accepts this language (any NFA is OK, but try to keep it as simple as possible).

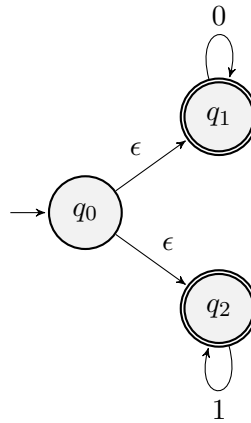
1. $1(010)^*0$

Answer (e.g.): 10 is in the language, 1 is not in the language.



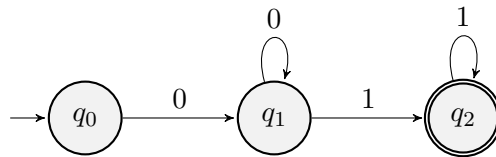
2. $0^* \cup 1^*$

Answer (e.g.): 0 is in the language, 01 is not in the language.



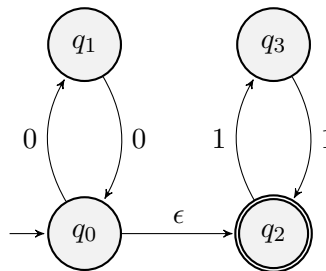
3. 0^+1^+

Answer (e.g.): 01 is in the language, 0 is not in the language.



4. $(00)^*(11)^*$

Answer (e.g.): 00 is in the language, 0 is not in the language.



Problem 3 (Closure Properties)

10 pts

1. Sipser 3rd Edition. Problem 1.31 on page 88.

Answer:

Let M be a DFA recognizing A , construct an NFA N recognizing $\mathcal{R}(A)$ by “reversing all the arrows” in M and adjusting the start and accept states. More specifically, the following transformations are made to M :

- Reverse all of the transitions in M , so if $\delta_M(q_i, \alpha) = q_j$ then $\delta_N(q_j, \alpha) = q_i$.
- Add a new state q_{new} that becomes the start state of N and add ϵ -transitions from q_{new} to all accept states from M .

- The only accepting state in N is the start state of M .

This construction can be seen visually below in Figure 1. It is clear from this construction that any computation path that ends in an accepting state in M can be followed identically in reverse order through N .

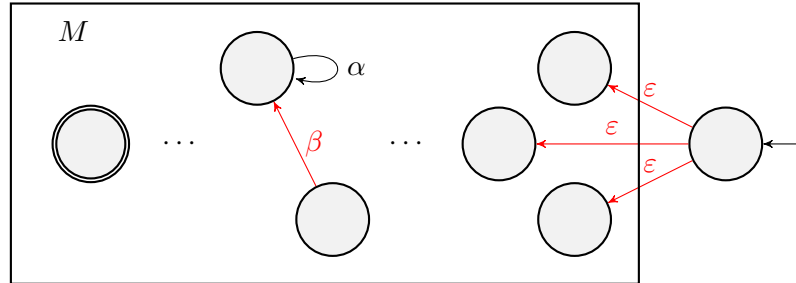


Figure 1: How to construct an NFA N with $L(N) = \mathcal{R}(L(M))$.

2. Sipser 3rd Edition. Problem 1.43 on page 90.

Answer: Let M be a DFA recognizing A , construct an NFA N recognizing $DROP-OUT(A)$ as follows:

- Create an additional copy of M called M' . The states of N consist of the disjoint union of the states in M and M' . All transition arrows from M and M' are preserved.
- The start state of N becomes the start state of M .
- For each transition $\delta_M(q_i, \alpha) = q_j$ in M add a new ε -transition $\delta_N(q_i, \varepsilon) = q'_j$ in N . That is, the new transition goes from a state in M to a state “one step forward” in M' .
- The accepting states in N are the accepting states from M' .

This construction can be seen in Figure 2 below. Intuitively this construction works by duplicating M into two “layers” which allow N to maintain the one additional bit of information indicating whether a deletion has occurred so far. The ε -transitions between layers allow N to “guess” when the deletion has occurred and what symbol was deleted, but after that computation proceeds normally in M' .

Problem 4 (More Closure Properties)

10 pts

1. (5 pts) Problem 1.70 in Sipser 3rd ed. page 93.

Answer: We’ll show that we can build A avoids B by performing operations on A and B that we know closure properties for. Define the operation

$$A \text{ has } B = \{w \in A : w \text{ contains a string from } B \text{ as a substring}\}.$$

First observe that we can write

$$A \text{ avoids } B = A \setminus (A \text{ has } B) = A \cap \overline{(A \text{ has } B)}.$$

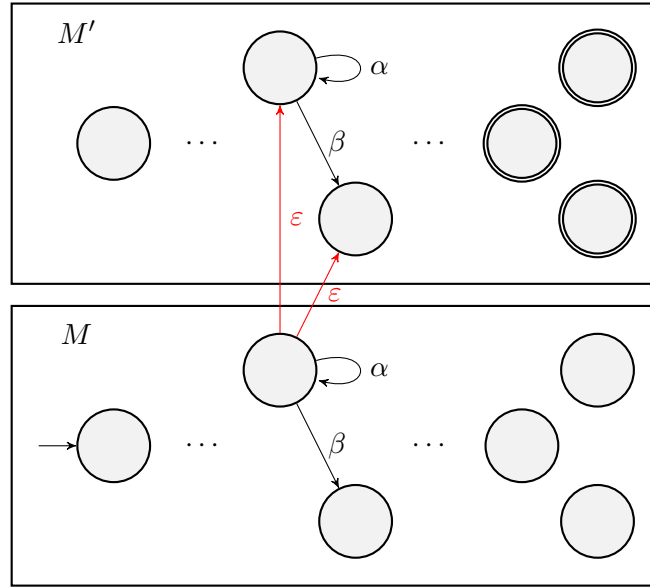


Figure 2: How to construct an NFA N with $L(N) = \text{DROP-OUT}(L(M))$.

Now observe that we can build $A \text{ has } B$ using only operations that we know closure properties for. We can write $A \text{ has } B = A \cap \Sigma^* B \Sigma^*$.

2. (5 pts) Does the following “converse closure” property hold?

If A, B are arbitrary languages such that $A \cup B$ is regular, then A and B is regular.

Either prove your answer or give a counterexample.

Answer:

No the property does not hold. There are several ways to construct counterexamples, and here is one: Let $A = \{0^n 1^n \mid n \geq 0\}$ and let $B = \{0, 1\}^*$. Then $A \cup B = \{0, 1\}^*$ is regular but A is non-regular. This gives a counterexample.

Problem 5 (Pumping Lemma)

10 pts

Use the pumping lemma to show that the following languages are *not regular*:

1. $L_1 = \{1^{n^3} : n \geq 0\}$.

Answer: Given the adversary’s choice p , choose $n = \max(1, p)$ and $w = 1^{n^3}$. Let x, y, z be the string chosen by the adversary such that $w = xyz$ with $|y| \geq 1$ and $|xy| \leq p$. We choose $i = 2$, then $xy^2z = 1^{n^3 + |y|}$. Clearly, $n^3 < n^3 + |y| < (n + 1)^3$ (because $n^3 + |y| \leq n^3 + n < (n + 1)^3$). Hence, $n^3 + |y|$ can’t be a cube and therefore $xy^2z \notin L_1$.

2. $L_2 = \{10^n 10^n : n \geq 0\}$.

Answer: Given the adversary’s choice p , choose $w = 10^p 10^p$. Let x, y, z be the strings chosen by the adversary such that $w = xyz$ and $|xy| \leq p$ and $|y| \geq 1$. Since $|xy| \leq p$, xy contains only symbols from the first half of w . There are two cases:

- If y contains a 1 (i.e. because $|x| = 0$), then choosing any $i \geq 2$ results in $xy^iz \notin L$ because it would contain more than two ones.
 - If y contains only zeros, then choosing $i = 2$ results in $xy^iz = 10^{p+|y|}10^p$ which is not in L because $p + |y| \neq p$.
3. $L_3 = \{ww^R : w \in \{0,1\}^*\}$ where w^R denotes the reversal of w . For example $(1000)^R = 0001$.
Answer: Given the adversary's choice p , choose $w = 0^p110^p$. Let x, y, z be the strings chosen by the adversary such that $w = xyz$ and $|xy| \leq p$ and $|y| \geq 1$. Since $|xy| \leq p$, y must contain only zeros. Now choose $i = 2$, then $xy^iz = 0^{p+|y|}110^p$ which is not in L because it is not a palindrome.